

MUHAMMAD MUTTAYAB

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🌐 linkedin.com/in/m-muttayab 🐙 github.com/Muttayab99 🌐 Portfolio

TECHNICAL SKILLS

- **Languages & Core:** [Python](#), C++, SQL, NoSQL, HTML/CSS, JavaScript.
- **Data Engineering & Cloud:** [AWS](#) (S3, SQS, EC2), [Azure Data Factory](#), [Databricks](#), [PySpark](#), [Kafka](#), ETL Pipelines.
- **AI & Machine Learning:** [PyTorch](#), [TensorFlow](#), [LangChain](#), [LangGraph](#), RAG, NLP (BERT, Llama), Scikit-learn, MCP.
- **Computer Vision:** [OpenCV](#), [SAM3](#), [PaddleOCR](#), [Google Vertex AI](#), Roboflow, Object Detection & Segmentation, Model Fine tuning.
- **Web & Visualization:** [FastAPI](#), Flask, Node.js, Power BI, Tableau, D3.js.

PROFESSIONAL EXPERIENCE

AI Engineer & Project Lead

Jan 2026 - Present

Neuralogic

Remote

- Led computer vision pipelines for automated construction cost estimation, fine-tuning [SAM3](#) (848M parameters) on a custom dataset using an [AWS EC2](#) instance with 80 GB VRAM, achieving precise element segmentation on architectural plans.
- Integrated [PaddleOCR](#) and [Google Vertex AI](#) to extract structured cost data from construction documents, reducing manual estimation effort significantly.
- Built and deployed production-grade [FastAPI](#) services to expose AI inference pipelines, including annotation management workflows and model prediction endpoints.
- Managed training data curation and model fine-tuning using [Roboflow](#), overseeing annotation quality control to ensure high-fidelity ground truth for CV models.
- Leading a second workstream as **Project Lead** for MEP (Mechanical, Electrical & Plumbing) plan estimation, coordinating team deliverables across model development and integration and architecting a **contract automation** module leveraging **GenAI** (LLM-based document understanding) to auto-generate and review construction contracts, streamlining end-to-end project documentation.

AI Engineer (Contract)

Oct 2025 - Dec 2025

SAynt AI

Remote

- Architected a multi-agent competitor analysis system in [Python](#) (Asyncio, Pydantic), integrating 15+ SEMrush endpoints with [AWS SQS](#) and [S3](#) for scalable message queuing and storage.
- Engineered an AI document processing API using [FastAPI](#), orchestrating multi-modal workflows with [Tesseract OCR](#), Google Vision, and [OpenAI GPT-4o](#) for automated data extraction.
- Designed a secure production architecture featuring [JWT authentication](#), HMAC security, and asynchronous background task management for high-throughput analytics.

Gen-AI Intern

July 2025 - Aug 2025

Addo AI

Remote

- Developed advanced **RAG** workflows using [LangChain](#) and [LangGraph](#), mastering **Prompt Engineering** and the Model Context Protocol (MCP).
- Integrated LLMs into business logic to automate complex decision-making processes, improving workflow efficiency for client solutions.

Data Engineering Intern

June 2024 - Aug 2024

Systems Limited

Lahore, Pakistan

- Optimized enterprise **ETL processes** using [Azure Data Factory](#) and [Azure Databricks](#), enhancing data throughput and reliability.
- Implemented distributed data transformation pipelines using [PySpark](#) to extract actionable insights from large-scale datasets.
- Developed interactive dashboards in **Power BI** to visualize KPIs and communicate data findings to executive stakeholders.

PROJECTS

Aerux - Intracranial Aneurysm Detection. Built a medical imaging system using a multi-task [ResNet50](#) pipeline in [PyTorch](#). Implemented robust preprocessing with [OpenCV](#) and SimpleITK (N4 bias correction, Sato vesselness). Developed an attention-based fusion model for 2.5D DICOM slices, leveraging mixed-precision training to optimize localization on the RSNA dataset.

MCP Orchestrator. Engineered an intelligent chatbot system with **FastAPI** and **Streamlit**, orchestrating workflows for video analysis and web scraping. Implemented NLP pipelines using **LangChain** and **Llama-4**, optimizing inference via Groq API and ensuring safety through sandboxed code execution.

Music Recommendation System. Developed a real-time recommendation platform using **Apache Kafka**, **MongoDB**, and **Flask**. Built audio feature extraction pipelines with **Librosa** (MFCCs) and utilized tree-based **ScaNN** for efficient vector retrieval, overcoming scalability limits of traditional KNN models.

DisasterReliefAI. Built an emergency resource allocation platform using **BERT transformers** for semantic context interpretation. Applied geospatial algorithms and Linear Sum Assignment Problem (LSAP) optimization to prioritize aid based on urgency and physical proximity.

VendorSys. Designed a full-stack vendor management platform with **Node.js** and a robust **SQL** backend. Implemented role-based access control (RBAC), cascading constraints, and automated trigger notifications to secure financial data and streamline procurement workflows.

Real-time Streaming Analytics. Implemented Apriori, PCY, and Page Rank algorithms for high-velocity data analysis. Utilized **Apache Kafka** for stream processing, enabling scalable frequent-itemset mining.

EDUCATION

Bachelors in Data Science	Aug 2022 - Present
FAST NUCES	Islamabad, Pakistan
• Coursework: Deep Learning, Info Security, PPIT, Computer Networking, Big Data Analytics.	
A - Levels	Aug 2019 - Aug 2021
Pak - Turk Maarif School & Colleges	Lahore, Pakistan

AWARDS & ACTIVITIES

- **Winner Hackathon**, HackFest FCS Charity Gala Islamabad.
- **Winner Debating Competition**, LPS Inter-school Competition.